Inorganic Cumulative Exam

February 23, 2001

Over the last 50 years, modern structural methods have revealed a remarkable diversity in molecular geometries. A number of "theories" or "collections of empirical rules" have been developed in an attempt to provide order to this diversity. The following questions can be answered using some of the more successful "theories". Answer the questions and indicate which of the theories you used to arrive at the answer.

1. Give the geometries of the following molecules.

   SF_4, SO_3, IF_3, XeF_4, AsH_3

2. Which of the following would be the most stable?

   $Fe(CO)_2(NO)_3$, $Fe(CO)_6$, $Fe(CO)_3$, $Fe(CO)_2(NO)_2$, $Fe(NO)_5$

3. Predict which ligand of the pair given would give the more stable complex with the indicated metal ion.

   Hg^{2+} with $(C_2H_5)_2O$ or $(C_2H_5)_2S$
   Co^{3+} with NH_3 or AsH_3
   Ni^0 with PCl_3 or NH_3

4. Draw the structures of:

   B_5H_9, $B_{12}H_{12}^{2-}$, B_2H_6, $B_{10}C_2H_{12}$.

5. Describe the structures of the following compounds, where C_4H_6 is butadiene and C_6H_6 is benzene.

   $Ni(CO)_2(C_4H_6)$
   $Fe(CO)_4(C_4H_6)$
   $Cr(C_6H_6)(CO)(C_4H_6)$
   $Co(C_4H_6)(CO)_2(NO)$